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Simultaneous electro-optic spectral switching and Q switching in a dual-wavelength Nd:YVO₄ laser based on an aperiodic optical superlattice lithium niobate

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We report the demonstration of an electro-optic (EO) switchable dual-wavelength (1064- and 1342-nm) Nd:YVO₄ laser based on an aperiodically poled lithium niobate (APPLN) chip whose domain structure is designed using aperiodic optical superlattice (AOS) technology. The APPLN works as a wavelength-dependent EO polarization-state controller in the polarization-dependent laser gain system to enable switching among multiple laser spectra simply by voltage control. When the APPLN device is driven by a voltage-pulse train modulating between a V_{HQ} (in which target laser lines obtain gain) and a V_{LQ} (in which laser lines are gain suppressed), the unique laser system can produce Q -switched laser pulses at dual wavelengths 1064 and 1342 nm, single wavelength 1064 nm, and single wavelength 1342 nm, as well as their non-phase-matched sum-frequency and second-harmonic generations at $V_{\text{HQ}} = 0, 267$, and 895 V, respectively. A laser can benefit from such a novel, to the best of our knowledge, simultaneous EO spectral switching and Q switching mechanisms to increase its processing speed and multiplexity for versatile applications. © 2022 Optica Publishing Group

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Fast switching among emission lines of a multi-wavelength laser can be an attractive technique in demand for various applications such as light detection and ranging (LiDAR), spectroscopy, chemical sensing, biomedical measurements, optical coherence tomography, and wavelength routing optical networks [1,2], in order to increase the processing capacity, speed, and multiplexity of the applied systems. Laser wavelength tuning has been realized in lasers having multiple transition bands or a broad gain bandwidth [3]. The most common wavelength-selective mechanism in lasers relies on the tuning of intracavity dispersive elements such as prisms, gratings, filters, or etalons. For example, several efforts have been made to demonstrate switchable multi-wavelength coherent light sources in (erbium-doped) fiber lasers based on the modulation of the birefringence or multimode conditions of the fiber Bragg gratings structured in the fiber-optic systems [4]. These spectral switching/selection

methods, however, have been all based on a relatively slow tuning mechanism using mechanics, which could hinder them from being a favorable solution for those aforementioned applications demanding a fast processing speed. Electro-optic (EO) effects have been widely employed to implement ultrafast (>GHz) optical modulators [5]. EO tunability has been realized in a diode laser with, however, a relatively limited tuning range due to a low EO-induced refractive index change in the diode structure [6]. To enhance the EO modulation effect, nonlinear-optic materials with a strong quadratic [$\chi^{(2)}$] nonlinearity can be employed. EO modulators have worked as an active laser Q switch for producing fast and high peak-power laser pulses [7]. However, the study of an EO mechanism in a laser to tune among its transition bands or resonant modes was hard to find; an EO switchable multi-wavelength laser system has not yet been demonstrated.

Lithium niobate (LN) could be the most iconic EO material because of its featured prominent Pockels effect, together with the access of its excellent nonlinear-optic, acousto-optic, and piezoelectric properties [8]. LN has been a versatile photonic substrate for various applications including the implementation of integrated coherent light sources [9]. Several of these intriguing results have been obtained based on the access of an off diagonal EO tensor element of LN, r_{51} , in which the power conversion between the two orthogonal [i.e., ordinary (o) and extraordinary (e)] polarization eigenmodes can be achieved through, e.g., a periodically poled LN (PPLN) (where the sign of r_{51} is spatially modulated) under the control of an electric field E_y along the crystallographic y axis [10]. Such an EO polarization-state controllable (PSC) device works as a polarization rotator and can play like an active narrowband notch filter when it operates in a polarization-dependent laser gain [such as a linearly (e - or π -) polarized Nd:YVO₄ laser] or nonlinear gain [such as a type-0 (eee) PPLN optical parametric oscillator (OPO)] system, in which novel integrated EO Q -switched lasers and fast spectral tunable or tailorable OPOs have been implemented [11].

More recently, we have first demonstrated an EO switchable multi-wavelength OPO based on a domain-engineered LN functioning both as a fast tunable notch-like filter and a multiline parametric gain medium [12]. In this study, we further build such a unique spectral-dependent EO PSC mechanism in an

LN chip using the aperiodic optical superlattice (AOS) technique [13] to achieve simultaneous EO spectral switching and Q switching in a dual-wavelength Nd:YVO₄ laser; to the best of our knowledge, this is the first laser system whose spectral and temporal characteristics can be fast switched simultaneously using simply the same EO effect.

To realize such a novel laser system, the core chip, which is an aperiodically poled LN (APPLN) (see below), is designed to integrate multiple EO PSC conditions to accomplish the following switching operations in a dual-wavelength ($\lambda_1 = 1064$ nm and $\lambda_2 = 1342$ nm) Nd:YVO₄ laser oscillating in e -polarized gain: (i) the laser can radiate at both lines, λ_1 and λ_2 , when the intracavity EO APPLN chip operates at the passive mode (i.e., at an applied voltage $V_{EO} = V_0 = 0$ V)—in this case, both λ_1 and λ_2 are e -polarized and get gain; (ii) the laser can radiate at one wavelength, λ_1 or λ_2 , when the APPLN chip operates at $V_{EO} = V_1$ (λ_1 is e -polarized and gets gain, while λ_2 is o -polarized and loses gain) or $V_{EO} = V_2$ (λ_1 is o -polarized and loses gain, while λ_2 is e -polarized and gets gain); (iii) the laser will cease when the APPLN chip operates at $V_{EO} = V_{LQ}$ (λ_1 and λ_2 are o -polarized and both lose gain). The above operations accomplish a CW, EO switchable dual-wavelength laser. Accordingly, and more interestingly, this laser can readily operate at a Q -switched pulse mode while performing the above EO spectral switching, i.e., the laser can emit Q -switched pulses at both λ_1 and λ_2 when the APPLN chip is driven by voltage pulses with a peak value (V_{HQ}) at V_0 (i.e., the “high Q ” state) and a bottom value (V_B) at V_{LQ} (i.e., the “low Q ” state); it can then be switched to emit laser pulses at λ_1 or λ_2 when the APPLN chip is driven by voltage pulses with $V_{HQ} = V_1$ or V_2 and $V_B = V_{LQ}$. All these switching conditions can be attainable thanks to the fact that the power conversion efficiency between the two polarization modes in the EO PSC device is a function of the driving field E_y ($\equiv V_{EO}/d$, where d is the electrode gap) and the laser wavelength λ_L given by [10]

$$\eta = \sin^2(|\kappa|L), \quad (1)$$

where L is the device length and $|\kappa| = \Im(\pi/\lambda_L)(n_o n_e)^{3/2}|r_{51}E_y|$, with $\Im$ being the Fourier reduction factor emerging from the characteristic QPM conversion scheme in a spatially varied domain polarity structure, and n_o and n_e being the refractive indices of the o - and e -polarization modes in the device. According to Eq. (1), for an e -polarized input laser, $\eta(V_{EO})$ will vary between 0 and 1 (in a sine-squared function) with the tuning of V_{EO} , where $\eta=0$ and 1 indicate the wave remains at an e -polarized state (a high-gain state in a Nd:YVO₄ laser) and is completely converted to an o -polarized state (a gain lost state in a Nd:YVO₄ laser) after passing through the device, respectively. The periodicity of such a voltage tuning curve can be determined by $\Im$ according to Eq. (1). Since $\Im$ is a function of the configured QPM superlattice and the wavelength, two voltage tuning curves with properly different periodicities for the two working wavelengths λ_1 and λ_2 can be created via the design of a special domain structure in the device to implement the four EO PSC conditions stated above [i.e., (i) both tuning curves reach $\eta=0$ (both remain at e mode) at $V_{EO} = V_0$, (ii) and (iii) the tuning curve for λ_1 (λ_2) reaches $\eta=0$ (at e mode) while the tuning curve for λ_2 (λ_1) reaches $\eta=1$ (at o mode) at $V_{EO} = V_1$ (V_2), and (iv) both tuning curves reach $\eta=1$ (at o mode) at $V_{EO} = V_{LQ}$] in a dual-wavelength Nd:YVO₄ laser. In contrast to our previous study of an EO switchable multiline OPO [12], in this work the four EO PSC conditions further accomplish the first observation

of simultaneous spectrum and Q (time domain) switching in a multi-wavelength laser system.

We adopted the AOS technique to find an optimized QPM domain structure and therefore $\Im(\lambda_1)$ and $\Im(\lambda_2)$ for creating the two voltage tuning curves for the device to accomplish the four EO PSC operations in an LN chip with the aid of the simulated annealing (SA) optimization method [14]. The AOS technique features high freedom and working efficiency in designing an optimized domain structure for a multi-wavelength QPM device with high conversion efficiency and target Fourier spectral fidelity in contrast with a periodically poled domain structure [15]. In our developed SA algorithm, an aperiodic domain structure constructed from a number (N) of unit domain building blocks with a thickness of Δx (usually much thinner than the coherence lengths of the conversion processes) will be derived via the search of an optimal sequence of domain polarities [corresponding to a sign (plus/minus) sequence of the relevant EO coefficient r_{51}] for the N blocks along the LN crystallographic x axis, resulting in an APPLN. In this algorithm, an objective function (OF) has been employed to be a measure of the achievement of the targeted “low-temperature” state (i.e., a desired solution) in the SA optimization process, given by

$$OF = \left(\sum_{i=1}^M w_{EO}(\lambda_i) |\xi_{EO,t}(\lambda_i, V_a) - \xi_{EO,c}(\lambda_i, V_a)| \right), \quad (2)$$

where M is the number of the working wavelengths (λ_i); $\xi_{EO,t}(\lambda_i, V_a)$ and $\xi_{EO,c}(\lambda_i, V_a)$ are, respectively, the target and calculated (based on the coupled-mode theory; Eq. (1) is a closed-form solution for a PPLN) power ratios of the e -polarized component of the wave with wavelength λ_i after making a round trip interaction with the EO APPLN device operating at V_a ; and $w_{EO}(\lambda_i)$ is the weighting factor for the conversion process with wavelength λ_i . The round trip interaction is considered in the calculation due to the EO APPLN chip having been designed to work in a laser cavity. In this study, $M = 2$, $\lambda_1 = 1064$ nm, and $\lambda_2 = 1342$ nm. The design temperature is 40°C. The OF helps construct an optimized APPLN whose operation is characterized by two voltage tuning curves (for λ_1 and λ_2) that not only meet the targeted four EO PSC conditions but also are formed with as low as possible operating voltages, V_1 , V_2 , and V_{LQ} . This is equivalent to finding the shortest achievable versions of two properly different periodicities for the two tuning curves, which has been pursued in the optimization process by determining appropriate power ratios $\xi_{EO,c}(1064$ nm) and $\xi_{EO,c}(1342$ nm) at a proper reference driving voltage V_a (=90 V set in this work) as the periodicity of the tuning curve is a function of $\xi_{EO,c}$ ($\propto \Im^2$, which is domain-structure dependent).

Figure 1(a) shows the calculated (round trip) voltage tuning curves of a 2-cm-long EO APPLN derived using the SA optimization algorithm for $\lambda_1 = 1064$ nm and $\lambda_2 = 1342$ nm waves initially polarized at the e mode at input. The inset shows the domain polarity structure of the calculated APPLN composed of unit domain width of $\Delta x = 5$ μ m. It clearly shows the two tuning curves of the device can support the four target EO PSC conditions (dotted lines in the figure); at 0 V (V_0), both two waves are polarized at the e mode [i.e., $\xi_{EO,c}(1064$ nm) = $\xi_{EO,c}(1342$ nm) = 1] as expected due to no effect of the polarization rotation (the passive mode); at 233 V (V_1), the 1064-nm wave remains at the e -polarized mode [$\xi_{EO,c}(1064$ nm) = 1], while most of the 1342-nm wave has been converted to o -polarized mode [$\xi_{EO,c}(1342$ nm) ~ 0.04]; at

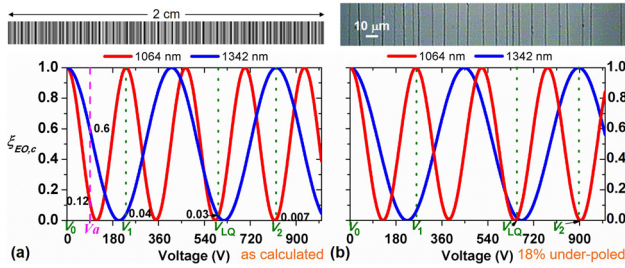


Fig. 1. Calculated round trip voltage tuning curves of the EO APPLN for $\lambda_1 = 1064$ nm and $\lambda_2 = 1342$ nm with (a) as-calculated, and (b) 18% under-poled domain structures. The insets show the domain polarity structures of the calculated (left) and fabricated (right) APPLN.

822 V (V_2), the 1342-nm wave remains at the e -polarized mode [$\xi_{EO,c}(1342 \text{ nm}) = 1$], while most of the 1064-nm wave has been converted to o -polarized mode [$\xi_{EO,c}(1064 \text{ nm}) \sim 0.007$]; at 595 V (V_{LQ}), both waves have been converted to o -polarized mode [$\xi_{EO,c}(1064 \text{ nm}) = \xi_{EO,c}(1342 \text{ nm}) \sim 0.03$]. These are the calculated optimum operating points for achieving the EO spectral switching as well as the EO Q switching in a dual-wavelength Nd:YVO₄ laser. In the actual experimental operation, however, a certain tolerance on the operating conditions (such as the switching voltages) can be expected as long as the gain of the lasing mode/line exceeds the laser threshold, and the gain of the suppressed mode/line is below the laser threshold. In Fig. 1(a), the dashed line shows the derived power ratios $\xi_{EO,c}(1064 \text{ nm}) \sim 0.12$ and $\xi_{EO,c}(1342 \text{ nm}) \sim 0.60$ at the reference voltage $V_a = 90$ V from the algorithm to obtain appropriate periodicities for the two tuning curves to enable the above four EO PSC operations in an APPLN chip.

The calculated domain structure was then translated into a photomask, via which the APPLN was fabricated in a 20-mm-long, 1-mm-wide, and 0.5-mm-thick z -cut LN chip using the standard photolithography and electric-field poling technique. The inset of Fig. 1(b) shows a microscopic image of a portion of the HF-etched + z surface of the fabricated APPLN chip, revealing excellent completeness of the domain grating pattern but with a non-uniformly under-poled error [16]. Such a domain poling error generally will lead to the change of the periodicities of the tuning curves, as shown in Fig. 1(b) for the result when an average under-poled error of 18% (deduced from the experimental results, see below) is applied to the APPLN domain structure in the calculation. It shows V_1 , V_2 , and V_{LQ} are now shifted to higher values of 259, 897, and 656 V, respectively. The end facets of the chip were then polished and coated with anti-reflection films for highly transmitting both 1064- and 1342-nm lines. Finally, the two y surfaces (spaced by 1 mm) of the APPLN chip were sputtered with NiCr electrodes for the E_y application. The device was then mounted on a temperature-controlled crystal oven and situate in a Nd:YVO₄ laser system to demonstrate the proposed intriguing EO spectral and Q switching effects, as schematically shown in Fig. 2. The a -cut, 0.5-at. % Nd:YVO₄ gain medium has a length of 7 mm and a cross section of $3 \times 3 \text{ mm}^2$. The pump source is an 809-nm fiber-coupled diode laser. The dual-wavelength oscillation of the laser is achieved in a collinear cavity scheme composed of three dielectric mirrors, designated M1, M2, and M3. The 1342- and 1064-nm lasers build up in the cavities formed by M1 and M2, and by M1 and M3, respectively. M1 is a 15-cm radius-of-curvature

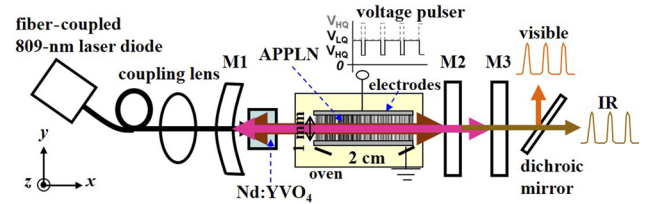


Fig. 2. Schematic arrangement of the EO spectrum and Q switchable dual-wavelength Nd:YVO₄ laser based on the APPLN chip.

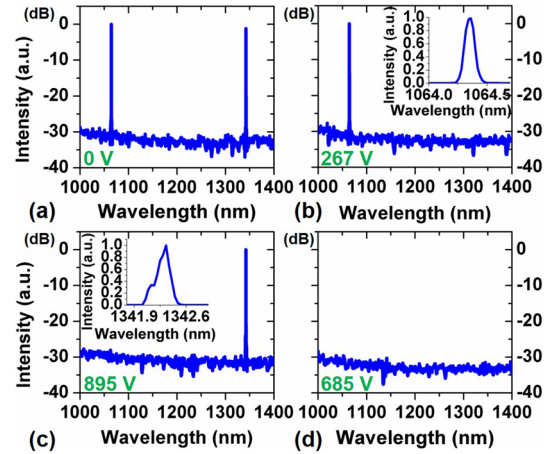


Fig. 3. Measured output spectra of the EO switchable (CW) Nd:YVO₄ laser when the APPLN is driven by (a) 0, (b) 267, (c) 895, and (d) 685 V under a diode pump power of ~ 7 W at 40°C. The insets in (b) and (c) show enlarged views (in linear scale) of the obtained spectral lines.

meniscus laser mirror having $>99.8\%$ reflectance at 1064 and 1342 nm and $\sim 95\%$ transmittance at 809 nm. M2 is a plane-plane laser mirror having $\sim 99\%$ reflectance at 1342 nm and $\sim 96.3\%$ transmittances at 1064. M3 is a plane-plane laser mirror having $\sim 62\%$ reflectance at 1064 nm and $\sim 99.3\%$ transmittance at 1342 nm. The cavity specification and configuration has been adopted mainly in consideration of the gain-competition balance between the two lines of different emission cross sections in Nd:YVO₄ [17].

Figure 3 shows the measured output spectra (in a dB scale) of the novel EO switchable (CW) Nd:YVO₄ laser system when the intracavity APPLN is driven by 0, 267, and 895 V, respectively, under a diode pump power of ~ 7 W at 40°C. It clearly shows EO spectral switching among dual 1064- and 1342-nm, single 1064-nm, and single 1342-nm lines has been demonstrated with the built laser system as designed but with higher working voltages than calculated [refer to Fig. 1(a)]. The corresponding slope efficiencies measured for the three switchable spectra were about 7.9%, 8.9%, and 6.5%, respectively. In the experiment, we have also observed the second-harmonic generation (SHG) and sum-frequency generation (SFG) of the two fundamental lines through the non-phase-matching wavelength conversion process in the APPLN, though with a low optical to optical conversion efficiency on the order of $10^{-4}\%$. The need for higher switching voltages in the experiment can be attributed to the APPLN domain poling error as discussed earlier; the result shown in Fig. 1(b) is a calculation best fitted to the experimental switching voltages, suggesting the fabricated APPLN has an

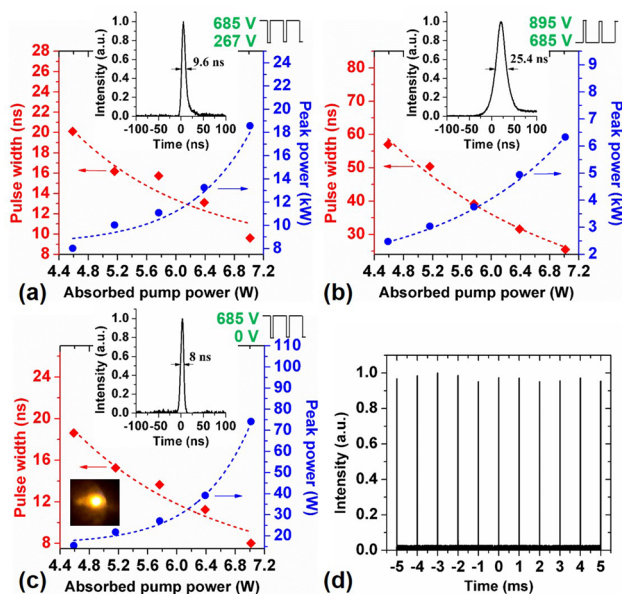


Fig. 4. Measured pulse energies and pulse widths of the generated Q -switched pulses from the laser system versus the diode pump power for the output of (a) single 1064-nm line (with $V_{\text{HQ}} = 267$ V and $V_{\text{LQ}} = 685$ V), (b) single 1342-nm line (with $V_{\text{HQ}} = 895$ V and $V_{\text{LQ}} = 685$ V), and (c) SFG (593.5 nm) of the dual lines (with $V_{\text{HQ}} = 0$ V and $V_{\text{LQ}} = 685$ V). (d) A typical pulse train of the generated Q -switched lasers from the system.

average under-poled error of $\sim 18\%$. Besides being tolerant to the fabrication error of the APPLN device, the laser system can also work over a certain range of the applied voltages as we have discussed earlier. According to our measurement, the EO switchable laser can have a voltage tolerance of ~ 30 V for each spectral switching operation.

Noticeably, when the APPLN is driven at 685 V, both laser lines are suppressed. It thus provides an excellent low Q -state for producing giant laser pulses for the dual 1064- and 1342-nm, the single 1064-nm, and single 1342-nm lines when making EO switching in the system between their respective CW operating (high Q -state) voltages 0, 267, and 895 V (V_{HQ}) and the low Q -state voltage 685 V (V_{LQ}). In operation, we drove the APPLN device with voltage pulses of an amplitude between V_{HQ} and V_{LQ} in a rise/fall time of ~ 30 ns and a duration of 300 ns at 1 kHz to repetitively Q switch the Nd:YVO₄ laser. Figures 4(a) and 4(b) show the measured pulse energies and pulse widths of the generated Q -switched pulses from the laser system as a function of the diode pump power for the output of single 1064-nm (with $V_{\text{HQ}} = 267$ V and $V_{\text{LQ}} = 685$ V) and single 1342-nm (with $V_{\text{HQ}} = 895$ V and $V_{\text{LQ}} = 685$ V) lines, respectively. The insets in the plots show the measured oscilloscope traces of the corresponding laser pulses at 7-W pump power. Again, in the measurement we have also observed the non-phase-matched SHG pulses at 532 and 671 nm and SFG pulses at 593.5 nm from the Q -switched dual-wavelength laser. Figure 4(c) shows the corresponding data measured for the SFG pulses produced when the system is driven with $V_{\text{HQ}} = 0$ V and $V_{\text{LQ}} = 685$ V, which provides the direct evidence of the synchronous generation of the dual IR pulses. A typical pulse train of the generated Q -switched lasers from the system was also recorded, as shown in Fig. 4(d). A peak-to-peak intensity fluctuation of $\sim 5\%$ was found.

Therefore, through the uniquely simultaneous EO spectral switching and Q switching mechanism, we have demonstrated the generation of six different laser spectra containing dual 1064- and 1342-nm, 1064-nm, 1342-nm, 593.5-nm, 532-nm, and 671-nm lines from this laser system simply by voltage control, workable over a wide diode pump power ranged from 4.6 to 7 W. Those non-phase-matched emissions can be further suppressed or enhanced through the domain engineering process. However, this will compromise the optimization for an efficient EO switching.

In conclusion, we have designed and implemented a novel APPLN chip based on the AOS technique to meet four EO PSC conditions targeted for conducting both fast spectral switching and Q switching in a dual-wavelength (1064- and 1342-nm) Nd:YVO₄ laser. The APPLN chip is the core device in the laser system that operates as an EO laser gain modulator described by two voltage and laser-line dependent polarization-state tuning curves, in which both and one of the dual lines can be fast switched electro-optically. When the APPLN device is driven by a fast (ns) voltage pulser with an amplitude switching between a V_{HQ} and $V_{\text{LQ}} = 685$ V at 1 kHz, Q -switched laser pulses at dual 1064 and 1342 nm, single 1064 nm, and single 1342 nm (and their SFG/SHG at 593, 532, and 671 nm) can be produced at $V_{\text{HQ}} = 0, 267$, and 895 V, respectively. Furthermore, such an EO switchable laser system can have a high operating tolerance in switching voltages ($\Delta V \sim 30$ V) and pump power (4.6–7 W).

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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